

AI Based Candidates Recommending System For a Company Vacancy

R25-040

Project Proposal Report

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DECLARATION OF CANDIDATE AND SUPERVISOR

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Effective stress management has been deemed an important skill in the professional setting, especially when stressful situations arise, such as during job interviews. The ability of a candidate to manage their stress greatly affects their communication, decision-making, and performance overall. This research examines the relationship between facial expressions and the stress response in interviews, focusing on important facial indicators such as eye movement, mouth shape, and eyebrow position. An intended outcome is to establish a systematic framework for observing these nonverbal signs as a way of measuring stress level variances. The research intends to develop an observable, documentable set of patterns of facial behaviors exhibited under controlled interview conditions so that there are measurable characteristics related to stress responses. The insights gained will elaborate on how stress is manifested through different facial expression variables and how this, in turn, affects interview performance. The study will also further develop a holistic perspective on soft skill assessment through the objective indicators of stress management, thus complementing an otherwise subjective interview protocol. The findings may lay the groundwork for the enhancing of recruitment practices through knowledge of candidates' nonverbal communication and emotional resilience in professional settings.

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1. INTRODUCTION

1.1 Background

Additionally, this has been noted that in training interviews, stress management does play an important role, as with any high-stress situation. A candidate's ability to manage their stress can dramatically affect their communication and decision-making abilities as well as their subsequent performances. Traditional interview regimes are reliant mainly on subjective judgments, which are, in fact, prone to bias or constraints on a person's stress level in a real-life situation. This marks the change to observations based on more objective assessments of stress responses, especially by way of behavioral and physiological signals, more specifically facial expressions.

Facial expressions are one of the key nonverbal behaviors that communicate information on the individual's emotional state at any given time. Any variation in eye movement, mouth shapes, or positions of the eyebrows is related to the stress response. From this point, scientists are embarking on developing far more viable methodologies in terms of stress detection by video expression analysis that are investigating the relationship between facial behavior and psychological states. Such technology will assess facial traits and offer insights from real-time responses of candidates during high-stress interviewing settings. The stress assessment through facial behavior is ultimately an objective tool that might, therefore, assist in improving selection processes by eradicating bias and increasing the accuracy of soft skill assessment.

1.2 Literature review

Recently, there has been a growing interest in the investigation of facial behavior analysis for stress detection, especially for real-time applications. A number of related studies have explored different methods of detecting the level of stress using facial expressions, deep learning, and multimodal approaches. Among the important contributions in this area, Rathore et al. (2021) discussed real-time stress detection using facial expression recognition. This research demonstrates how CNN can accurately classify stress-related emotions like fear and anxiety. The study indicates the promise of deep learning in emotion recognition but lacks a specific application for stress detection in high-pressure scenarios like job interviews [1].

Similarly, Zhang et al. (2022) propose a multimodal approach to stress detection, including facial expressions, verbal tone, and physiological signals. Their findings show that increasing the diversity of data sources significantly enhances the accuracy of stress detection. However, their work does not target interview scenarios but relies mostly on physiological sensors, which are unlikely to be adopted in real-world, non-invasive applications [2].

Another interesting work is that by Thomas et al., 2023, about real-time emotion recognition based on quantitative facial motion analysis. In this work, micro-expression tracking and minor face movements are undertaken to characterize an emotional state. Although the study provides useful insight into facial motion-based emotion recognition, it has not directly spoken to the question of detecting stress in high-pressure scenarios such as job interviews [3].

Smith and Patel [4], 2023, present a work on detecting stress in autistic persons during simulated job interviews through facial expression-based approaches fused with physiological sensors. It brings up issues of people getting stressed out in structured interview situations. As effective this could be in

detecting the stress in interviews using AI methodologies, dependency on external physiological sensors makes them less ideal to employ in the current nonintrusive webcam approach.

These works collectively underline the successes of stress detection using AI and face analysis. However, real-time non-obtrusive stress classification systems are lacking for job interview scenarios. This paper, therefore, seeks to bridge this gap by designing a web camera-based AI model for the detection and prediction of applicants' level of stress in an interview process to better understand behavioral responses.

1.3 Research Gap

Existing studies primarily concentrate on multimodal stress detection through wearable sensors or physiological signals, which makes them less suitable for real-world interview settings [1], [2]. There is a noticeable gap in research regarding a real-time, webcam-based stress detection model specifically designed for high-pressure interviews [3], [4].

Specifically:

1. Lack of Real-Time Stress Analysis in Interview Contexts

- Research studies examine generic stress detection yet they exclude stress-related evaluation of job interview situations.

2. Limited Webcam-Based Non-Intrusive Stress Detection

- The current research uses multimodal analysis with physiological devices that prove incompatible for genuine interview scenarios.

3. Absence of Behavioral Analysis under High-Pressure Scenarios

- The existing analysis methods fail to explore comprehensive changes in candidate facial expressions during stressful job interview sessions.

4. Limited Machine Learning Models for Interview Stress Prediction

- Review of AI-based predictive models for interview stress classification remains minimal because no specific research exists about these models for stress assessment.

Contribution of the Proposed Research

Our research aims to address these gaps by:

Webcam-Based Stress Analysis for Interview Candidates

- The stress analysis system is more available and allows the absence of other external physiology sensors

Integration of Machine Learning Models for Stress Level Prediction

- Advanced deep learning models will be designed to determine the stress levels during the active facial movement.

Real-Time Behavioral Analysis under High-Pressure Interview Scenarios

- The system will analyze and profile interview candidate's facial expressions and micro-expressions in order to detect stressors in real time.

Improved Interview Candidate Assessment for Employers

- The system provides recruiters with insights into candidates' stress levels, enabling better decision-making in high-stakes interviews.

1.4 Research Problem

The research addresses the following key problems:

1. **A machine learning model requires design specs to detect facial expressions from webcams to predict stress levels of candidates during interviews?**
2. **need to determine which specific facial expression elements including eye movements along with eyebrow placements and mouth structures help accurately measure candidate interview stress?**
3. **Real-time stress detection technology needs to be combined with AI interview systems for measuring candidate performance under demanding interview conditions?**

The research problems establish a systematic strategy to improve candidate assessment techniques by resolving previously detected research deficits.

2. OBJECTIVES

2.1 Main Objective

A system needs development which assesses interview candidates' stress levels by analyzing their facial behavior with computer vision algorithms and machine learning techniques.

2.2 Specific Objectives

- The development focuses on selecting distinct facial indications of interview-related stress which include eye behaviors and mouth shapes and eyebrow positions.
- A machine learning model needs development for analyzing facial expression features and stress level prediction accuracy.
- Testing of the stress prediction system involves collecting webcam data which will be used to validate the model performance.
- The stress detection model needs implementation into a computerized assessment system for interviewing job seekers under demanding situations.

3. METHODOLOGY

The research proposes building an in-time stress assessment framework based on facial behavior monitoring for interview candidates to deliver precise analysis of their stress levels. The system starts its operation by acquiring video streams from webcams which operate without requiring any wearables. Video processing takes place frame-by-frame to extract facial expression features starting from eye movements and continuing with mouth shape and eyebrow positions up to micro-expressions because these features link to stress responses.

To analyze these expressions effectively, the system employs computer vision techniques for facial landmark detection, allowing it to track subtle facial changes over time. Machine learning models analyze video-based features after they are extracted from stress-related expression datasets that have received labels. The approach applies deep learning methods specifically Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for making instant stress assessments of candidates. After stress prediction is accomplished the system produces an assessment report or live alerts which showcase stress trend data from the interview session to the company. Through integration with an AI-based interview platform the notification system generates professional insights about candidates' responses to pressure during evaluations. The model operates through continuous performance upgrades using new data for better precision and adaptability with different candidate profiles. The evaluation methodology bases hiring decisions on collected data which supports fair assessment of candidate stress tolerance and identifies qualified individuals for high-pressure roles.

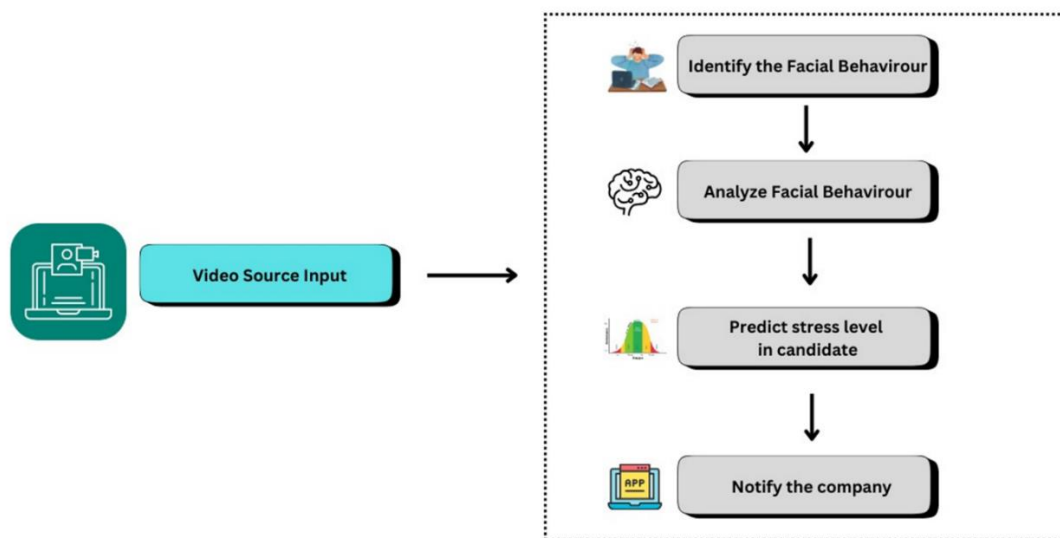


Figure 3. 1 component overview diagram

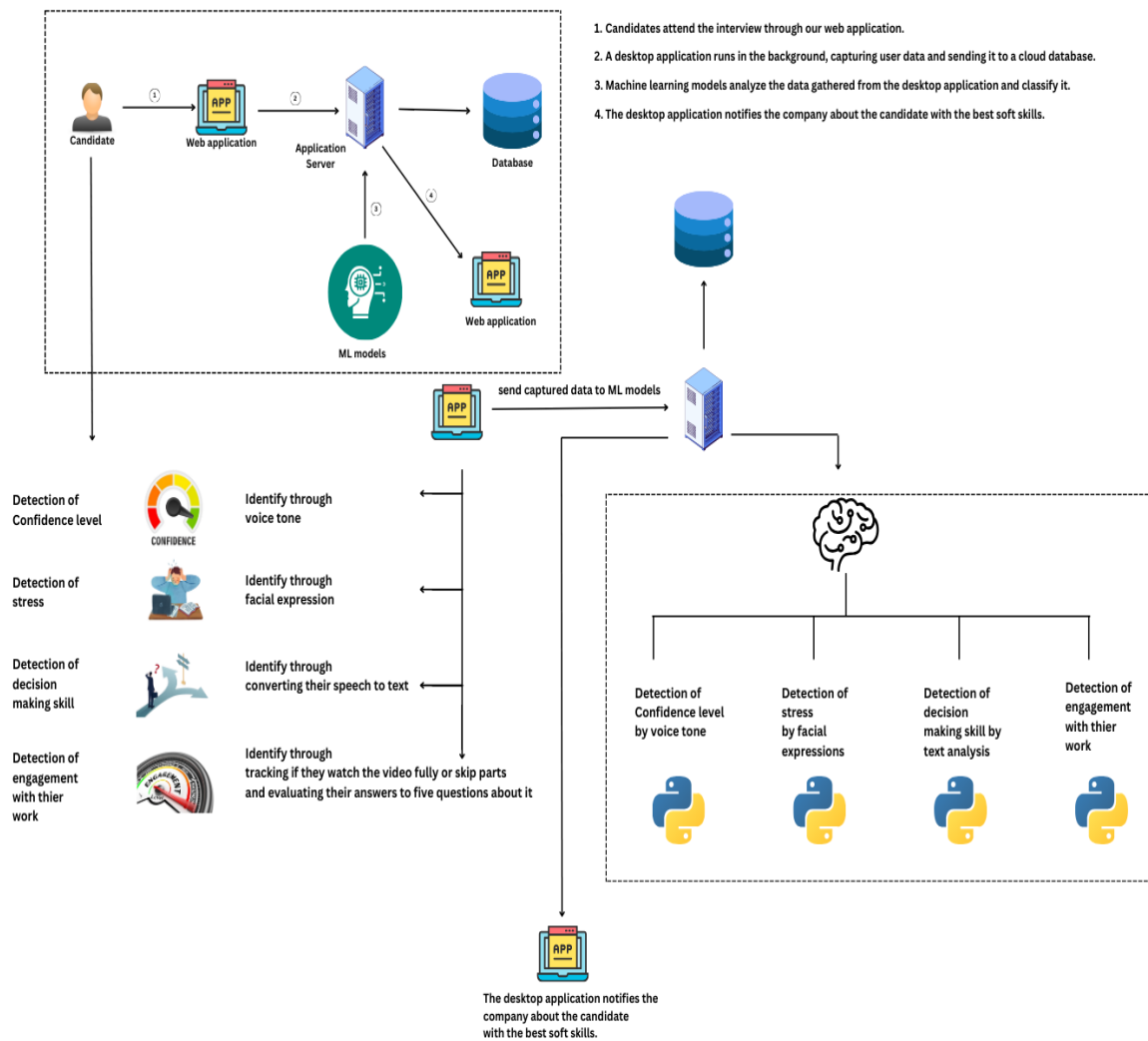


Figure 3. 2 overview System diagram

4. PROJECT REQUIREMENTS

4.1 Functional requirements

- **Real-Time facial expression analysis:** The system must process the candidate's facial expressions via a web camera in real-time. In addition to this, a conversion of data should come.
- **Stress-Level Prediction:** Firstly, the system is assumed to predict the stress level of a candidate by processing facial cues such as the movement patterns of the eyes, the position of the eyebrows, the shape of the mouth, as well as specific body signals.
- **Machine Learning Integration:** To achieve a facial emotion recognition system is to use available technology like deep learning which can be used for learning purposes of identifying humans? The facial expressions and then the resulting information about them, i.e., the degree of stress.
- **Facial Feature Detection:** System must be able to select certain facial features, such as eye or lip movements, that can be pointed out as stressful expression signposted, ("blue" is a color). Furthermore, stress becomes the major factor that is responsible for the appearance of the facial qualities mentioned.

4.2 Non-Functional requirements

- **Performance:** For real-time operation the system should be capable of detecting and analyzing the facial data processing results in less than 2 seconds.
- **Accuracy:** The success of the stress detection model is dependent upon its ability to achieve at least 90% accuracy of stress levels prediction through facial expressions.
- **Scalability:** The system has to be able to do multi-candidate analysis of participants in group interviews at the same time without performance getting lowered.
- **Usability:** The user interface of the system needs to offer user experience which is more oriented towards the perspectives of the users, namely recruiters, who could easily understand both software and system reports when being.

4.3 User requirements

For Candidates:

- **Non-Intrusive Experience:** The interviewee will not need any additional equipment in their interview experience, because the system itself is non-intrusive.
- **Clear Instructions:** The system gives step-by-step instructions about the camera angle to the applicants who use the system for interviews.
- **Privacy Assurance:** It is important for the candidates to get an answer if their facial data, obtained during the application of the system, is stored privately or not, because it is used to the interview for stress assessment needs and will not be stored or distributed unless the user gives the approval.

For Administrators:

- **Real-Time Monitoring:** Admins must have to check stress levels of interview candidates in real
- **User-Friendly Interface:** The administrator should get a system that is simple to use and that shows a stress prediction on one side and a trend in stress level of a candidate on the other.

4.4 System requirements

Hardware Requirements:

- **Webcam:** A web camera is one of the system requirements and it has to be of high definition (minimum 1080p) for capturing facial reactions with good detail.
- **Processing Unit:** The Intel i7 The system needs a PC with at least an Intel i7 CPU and a graphics card that is no lesser than NVIDIA GTX 1060 to run it in real time and run machine learning model calculation.
- **Memory:** RAM with at least 16GB
- **Storage:** SSD storage with capacity of 500GB at the minimum for the training datasets, logs, as well as temporary video files.

Software Requirements:

- **Operating System:** Windows, macOS, or Linux.
- **Facial Recognition Libraries:** OpenCV, Dlib, or MediaPipe are not only used for face analysis, but also in the case of feature extraction.
- **Machine Learning Frameworks:** The stress detection model requires TensorFlow or PyTorch or Keras as machine learning frameworks when building its deployment.
- **Programming Languages:** To build its system framework and system interfaces, Python programming language will be needed.
- **Database:** MySQL or MongoDB databases are required by the system for storing system logs as well as user data together with stress analysis outcomes.
- **User Interface:** User-friendly desktop or web applications are developed using programming frameworks such as React and Angular or Electron for improved usability.

Network Requirements:

- **Internet Connectivity:** Using screened and secured network access through the Internet with at least a 5 Mbps upload speed is the first determining factor to be able to stream in real-time the cloud-based setup.
- **Latency:** The system execution will be more efficient if it has the least latency (less than 100ms) needed for seamless stress detection and feedback.
- **Security:** The system implements HTTPS and SSL/TLS protocols to guarantee security for candidate data when they stream videos and undergo analysis.

4.4 Expected Test Cases

Functional Test Cases

Test case ID	Description	Input	Expected Output	Pass/ Fail Criteria
FTC01	Verify real-time facial expression capture using a webcam	Candidate sits in front of the webcam during an interview.	System captures video and processes facial expressions in real-time.	Video is captured without lag, and facial features are detected accurately.
FTC02	Validate detection of key facial features (eye movements, mouth shape, eyebrows)	Candidate performs specific facial expressions (frown, smile)	The system detects and tracks eye movements, mouth shape, and eyebrow positions.	Facial features are correctly identified and tracked in real-time.
FTC03	Test stress level prediction based on facial expressions.	Candidate exhibits stress-related facial expressions (furrowed brows).	The system predicts stress level (low, medium, high).	Stress level prediction matches the expected output based on facial expressions.
FTC04	Verify integration of the machine learning model for stress detection.	Pre-recorded video of a candidate under stress is fed into the system.	System processes the video and outputs stress levels.	Stress levels are predicted accurately based on the model's training.
FTC05	Test real-time notifications for recruiters.	Candidate exhibits high stress during the interview	Recruiters receive a real-time notification indicating high stress levels	Notification is sent promptly and accurately reflects the candidate's stress level

Non-Functional Test Cases

Test case ID	Description	Input	Expected Output	Pass/ Fail Criteria
NFTC-01	Verify system response time for real-time stress detection.	Candidate exhibits stress-related facial expressions.	The system predicts stress level within 2 seconds.	Stress level is predicted within the specified response time
NFTC-02	Test system accuracy in stress level prediction	Pre-recorded videos with known stress levels are fed into the system.	System achieves $\geq 90\%$ accuracy in stress level prediction.	Accuracy meets or exceeds the 90% threshold.
NFTC-03	Validate system scalability with multiple candidates.	5-10 candidates participate in a group interview	System processes facial expressions of all candidates without performance issues	No lag or degradation in performance with multiple candidates.

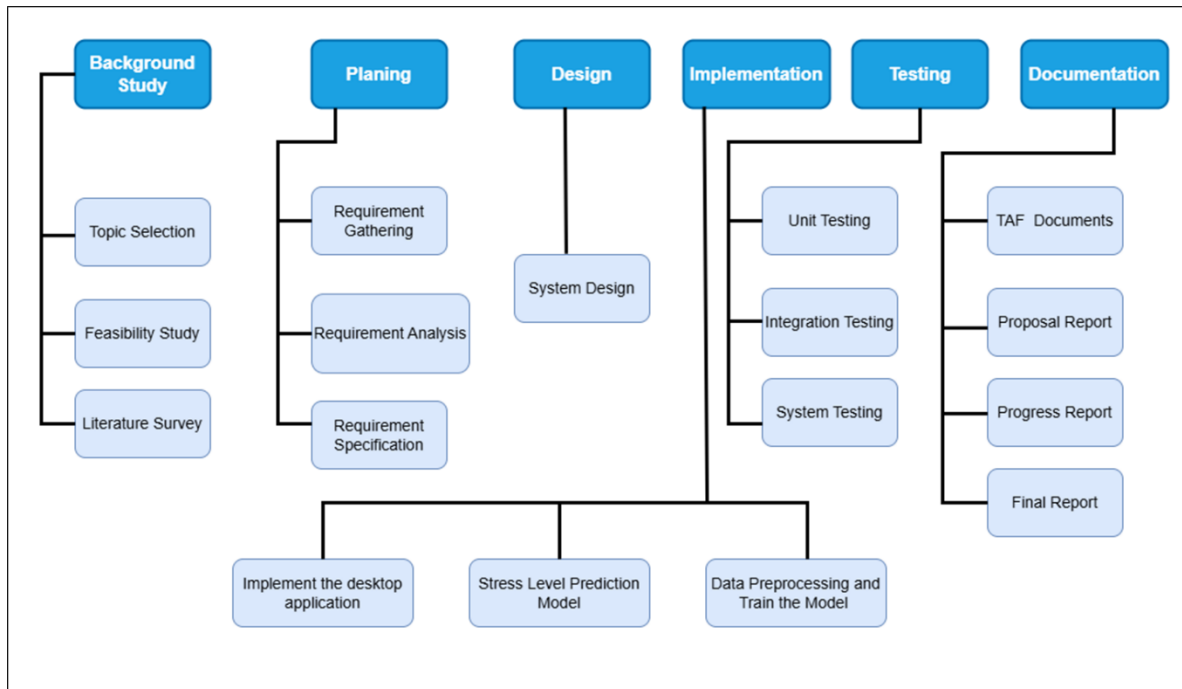
Usability Test Cases

Test case ID	Description	Input	Expected Output	Pass/ Fail Criteria
UTC-01	Test ease of setting up for recruiters.	Recruiter sets up the system for an interview	System is ready for use within 10 minutes	Recruiters can set up the system without technical assistance
UTC-02	Test candidate experience during the interview.	Candidates participate in an interview using the system.	Candidates do not feel uncomfortable or distracted by the system.	Candidate feedback indicates a non-intrusive and seamless experience.

Edge Case Test Cases

Test case ID	Description	Input	Expected Output	Pass/ Fail Criteria
ETC-01	Test system behavior with no facial expressions (e.g., candidate looks away).	Candidate looks away from the webcam for an extended period.	System detects absence of facial expressions and logs it appropriately	System handles the scenario gracefully without errors.
ETC-02	Test system behavior with extreme facial expressions (e.g., exaggerated smiles).	Candidate exhibits exaggerated facial expressions	The system accurately distinguishes between normal and exaggerated expressions.	Stress level predictions remain accurate despite exaggerated expressions

5. WORK BREAKDOWN STRUCTURE



6. GANTT CHART

	2024	2025											
Tasks	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Project Initiation													
Finding the topic													
Finding the Client													
Finding a Registration													
Project Planning													
Identifying the Problem													
Feasibility Study													
TAF Documentation													
TAF Evaluation													
Project Proposal Presentation Preparation													
Project Proposal Evaluation													
Project Proposal Report Documentation													
Project Proposal Report Submission													
Implementation													
Collecting Data													
Component 1													
Component 2													
Component 3													
Component 4													
Progress Presentation													
Integration													
Research Paper Publication													
Testing													
Unit Testing													
Component Testing													
Integration Testing													
Progress Presentation 2													
Deployment													
Deployment to Production													
Final Report Presentation													
Final Presentation and Viva													
Final Report Submission													

7. Budget & Budget Justification

Category	Item	Estimated Cost (LKR)	Justification
Hardware	Camera(web)	20000	Required for capturing clear facial expressions during interviews.
	16GB RAM	14000	Ensures smooth operation of the system during real-time analysis.
Software	Facial recognition libraries (OpenCV, Dlib)	0	Open-source tools for facial landmark detection and feature extraction.
	Machine learning frameworks (TensorFlow, PyTorch)	0	Open-source frameworks for developing and training the stress detection model.
	Database (MySQL/MongoDB)	0	Open-source database for storing system logs and user data.
	UI development frameworks (React, Angular)	0	Open-source frameworks for building the user interface.
Total		18000	

Budget Justification

1. Hardware Costs:

- The high-resolution webcam is a component of the system that captures and processes human face movements in real time, "economic spending" whereas, these parts make the system work without delays.
- Having 16GB RAM and 500GB SSD is essential for the management of large data sets and the assurance of the smooth functioning of the system.

2. Software Costs:

- Tools such as those found in OpenCV, TensorFlow, and React are the ones used in keeping costs low while ensuring use of the most powerful and the most widely adopted technologies.
- Using licensed datasets is essential in the

8. REFERENCES

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- [4] L. C. Smith and D. N. Patel, "Stress Detection of Autistic Adults during Simulated Job Interviews," *Proceedings of the ACM Digital Library*, vol. 22, no. 7, pp. 120-129, Oct. 2023.

9. APPENDICES

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